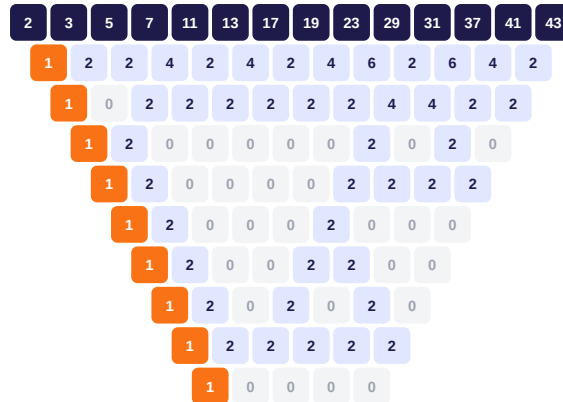


Gilbreath's Conjecture

List the primes. Subtract neighbors to make a new row. Keep going. Every new row starts with 1.



The primes on top, then rows of differences. The first number of every row is 1, in orange.

TRY IT

Primes: 2, 3, 5, 7, 11, 13, 17...

Differences: 1, 2, 2, 4, 2, 4...

Next row: 1, 0, 2, 2, 2...

Next row: 1, 2, 0, 0...

WHAT MATHEMATICIANS KNOW

- Norman Gilbreath noticed it in 1958 while doodling on a napkin. François Proth had claimed a proof in 1878, but it was flawed.
- Andrew Odlyzko checked it in 1993 for the first 340 billion primes.
- Some mathematicians suspect it has less to do with primes than with any list that starts with 2, keeps growing, and has small, irregular gaps.

WHY IT'S STILL UNSOLVED

The pattern depends on the gaps between primes staying fairly small and irregular. Proving that takes more knowledge about prime gaps than anyone currently has.

TRY THIS WITH STUDENTS

- Build the triangle together on the board. It needs only subtraction, so every student can contribute a number.
- Ask: Why do the rows fill up with 0s and 2s? What does a row of 0s and 2s turn into?
- Try a different starting list, like 1, 2, 4, 8, 16, or the class's birthdays in order. Does “always 1” survive? That tests whether the primes matter.

WORDS TO KNOW

Difference the larger number minus the smaller

Neighbors numbers next to each other in the list

Online, with the other seven problems: ohiomathteacher.github.io/applet-library/open-problems/gilbreath.html · The picture was computed from the real numbers.